

Harshwardhan Saini

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Objective

Computer Science undergraduate specializing in AI-driven systems and scalable backend architecture. Experienced in building multi-agent platforms, real-time data pipelines, and production-grade cloud applications across AWS and GCP.

Experience

DT Intern

GE Appliances (Haier) | Jan 2026 – Present

- Built multiple AI conversational agents (ADK, Dialogflow CX) enabling natural language querying of manufacturing data, reducing reliance on manual dashboards.
- Developed backend APIs, cloud infrastructure, and CI/CD pipelines enabling contractor-developed multi-agent orchestration and MCP integrations.
- Delivered end-to-end features across Angular frontend, AI agents (Agent Engine), MCP (Cloud Run), and serverless APIs (AWS Lambda & Google Cloud Functions) and built reusable Angular components for the ADK agent UI, later packaged as a shared library adopted across internal applications.
- Engineered an MCP-based system enabling natural language access to BigQuery, with user-configurable agent prompts via OpenWebUI.
- Built the foundation for autonomous AI agent workflows including Cypress test case generation and execution for Angular applications.
- Automated application deployments and infrastructure provisioning using GitHub Actions, Jenkins, and Terraform in cloud environments.

Projects

National Cyber Threat Intelligence Hub (Team project)

- Developed a microservices-based cyber threat intelligence system with real-time log processing using Kafka and honeypots.
- Owned backend architecture(Node.js (Express) & Django): designed and implemented core microservices, authentication system, and API gateway, integrating them into a unified platform for real-time threat ingestion, processing, and monitoring.
- Built ETL pipelines in Node.js (Express) for near real-time ingestion and processing of attack logs.
- Integrated backend services with a React dashboard using WebSockets for real-time monitoring.
- Deployed the system on Minikube and implemented CI/CD pipelines using Jenkins.

Sage AI (Personal Project) - [link](#)

- Built a full-stack AI chatbot using MERN, FastAPI, Redis, and MongoDB with vector database support for semantic retrieval.
- Implemented JWT and OAuth2-based authentication along with chat management and reusable prompt system.
- Developed features such as interactive quizzes and personalized feedback using LLM APIs.
- Deployed the application using free-tier services (Vercel, Render, MongoDB Atlas, Upstash Redis), designing a cost-efficient architecture for scalable, resource-constrained environments.
- Improved chatbot experience by enabling structured conversations and efficient prompt reuse.

Education

NIIT University | Btech in Computer Science | 2022-26

Technical Skills

- **Programming Languages:** Python, JavaScript, TypeScript, Go
- **Frontend:** HTML5, CSS3, TailwindCSS, React.js, Next.js, Angular
- **Backend & APIs:** Node.js, Express.js, Flask, FastAPI, Django
- **Agentic AI & GenAI:** LLM APIs, Prompt Engineering, Dialogflow CX, ADK, AI Agent Development, MCP
- **Databases (SQL & NoSQL):** MySQL, MongoDB, Redis, Aurora
- **Cloud & DevOps:** AWS, GCP, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, Git
- **Systems & Architecture:** Microservices, Kafka, WebSockets, Multi-Agent Systems